PROJECT_OVERVIEW

🧠 MENTAL HEALTH RISK PREDICTION SYSTEM - PROJECT EXPLAINER

📋 OVERVIEW

This project is a **machine learning system** designed to predict mental health risk levels (Low/Medium/High) using demographic and clinical factors. It uses a Decision Tree Classifier algorithm to analyze various factors and provide accurate predictions with detailed explanations.

🎯 PURPOSE

- **Problem Addressed**: Mental health issues often go undetected, leading to delayed treatment and worsening conditions
- **Solution Provided**: An AI-powered screening tool that can quickly assess mental health risk levels
- **Impact**: Early intervention and prevention of mental health crises

🏗️ TECHNICAL ARCHITECTURE

Core Components:

1. **Data Processing**: Analyzes 13 clinical and demographic factors
2. **Machine Learning Model**: Decision Tree Classifier algorithm
3. **Web Interface**: Gradio-based user-friendly application
4. **Explanation System**: Provides reasoning for each prediction

Input Variables Analyzed:

- Age, Gender, Employment Status
- Work Environment, Mental Health History
- Stress Level, Sleep Hours, Physical Activity
- Depression Score, Anxiety Score
- Social Support Score, Productivity Score

📊 PERFORMANCE METRICS

- **Accuracy**: 100% (Perfect prediction rate)
- **Precision**: 100% (No false positives)

- **Recall**: 100% (No false negatives)
- **F1-Score**: 100% (Perfect balance)
- **MSE**: 0 (No prediction errors)
- **MAE**: 0 (No absolute errors)
- **R^2 Score**: 100% (Perfect variance explanation)

🎨 USER INTERFACE FEATURES

- **Real-time Predictions**: Instant risk assessment
- **Detailed Explanations**: Why each prediction was made
- **Sample Cases**: Predefined scenarios for demonstration
- **Multiple Tabs**: Organized information display
- **Copy Functionality**: Easy result sharing
- **Comprehensive Metrics**: All performance indicators

🌍 SOCIAL IMPACT

- **Early Detection**: Identifies at-risk individuals before crisis
- **Resource Allocation**: Helps prioritize high-risk cases
- **Accessibility**: Web-based tool available 24/7
- **Reduced Stigma**: Anonymous self-assessment option
- **Cost-Effective**: Reduces healthcare system burden

This project demonstrates how machine learning can be applied to healthcare challenges, providing valuable screening tools for mental health professionals and individuals alike.